ITA0329 - Mobile Computing
CO1 - AT1 - Concept Mapping

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FREQUENCY REUSE

Requires
Frequency planning
- Assigns frequencies to cells
- Reduces co-channel interference
- Ensure efficient reuse pattern

May cause
Co-channel interference
- occurs in cell using the same frequency
- Reduces signal quality
- Affects call drop rate

can cause
Adjacent-channel interference
- occurs in nearby frequency
- causes signal leakage

Increases
Network capacity
- Allows more users
- Increases no. of channels
- Improves spectrum utilization

Interference Reduction techniques

Frequency planning
- Proper channel allocation
- Reuse distance

Power Control
- Adjusts transmission power
- Minimizes interfere

Sectoring / directional Antennas
- Focuses signal in a direction
- Reduces overlap

Guard Bands and filtering
- using space between channels
- Filters Unwanted signal

Better Cellular Network performance